

Phase 0.2 — Kepler's Equation by Hand

Type: Tool drill, no mission narrative. **Target time:** ~20 minutes per example once the iteration pattern clicks — budget longer the first time through. **Prerequisite:** 0.0 Slide rule basics, 0.1 Trig/log table interpolation.

This is the drill the handoff notes call out specifically: solving Kepler's equation by hand, iteratively, with a slide rule and trig tables, is something nobody in the group has actually done — even people with graduate-level orbital mechanics training almost always learned it as code, not as a hand procedure. This is where that changes.

The equation

Kepler's equation relates mean anomaly (time-like, uniform) to eccentric anomaly (geometric, on the auxiliary circle):

$$M = E - e \cdot \sin(E) \quad (\text{radians})$$

Worked in **degrees** (the practical choice with a degree-based trig table), this becomes:

$$M = E - k \cdot e \cdot \sin(E) \quad \text{where } k = 180/\pi \approx 57.29578$$

There's no closed-form solution for E — it's solved iteratively. Newton-Raphson on this equation, in degrees:

$$\begin{aligned} g(E) &= E - k \cdot e \cdot \sin(E) - M \\ g'(E) &= 1 - e \cdot \cos(E) \\ E_{\text{new}} &= E - g(E)/g'(E) \end{aligned}$$

Watch the units carefully here — this is the single most common place to introduce an error. $g'(E) = 1 - e \cdot \cos(E)$ has **no** factor of k in it, even though $g(E)$ does. That's not a typo: differentiating $\sin(E)$ with respect to E when E is expressed in degrees pulls in a factor of $\pi/180$ from the chain rule, which exactly cancels the k in the sine term. If you find yourself writing $1 - k \cdot e \cdot \cos(E)$, stop — that's the mistake to check for first.

A good starting guess: **$E_0 = M$** . For higher eccentricity, $E_0 = M + k \cdot e \cdot \sin(M)$ converges faster (it's literally the first pass of simple successive substitution) — worth trying on Example 2 below if you want to compare iteration counts.

Reference material

Trig table: use the 0.1 excerpt/technique (interpolate sin and cos to at least 5 decimal places — errors here feed directly into $g(E)$ and $g'(E)$).

Period-authentic note: purpose-built tables existed historically for solving Kepler's equation directly by lookup rather than iteration (e.g. tables of E as a function of M and e , in the tradition of Bauschinger's orbit-determination tables). Sourcing and linking a genuine period edition of one of these is flagged as a follow-up for reference/tables.md — not yet done, so this drill uses direct Newton-Raphson iteration instead, which is itself a legitimate period-appropriate method, just more laborious than a purpose-built lookup table would have been.

Example 1 — Low eccentricity

Given: $e = 0.100$, $M = 30.000^\circ$

Iterate $E_{\text{new}} = E - g(E)/g'(E)$ starting from $E_0 = M$, until $|\Delta E|$ is comfortably below your target precision (aim for $< 0.01^\circ$).

Example 2 — Higher eccentricity

Given: $e = 0.400$, $M = 45.000^\circ$

Same procedure. Expect more iterations and a bigger first correction — this is exactly why higher-eccentricity orbits were the harder case for real human computers, and why a smarter initial guess mattered more as e grew.

Once E converges for either example, get the true anomaly ν :

$$\tan(\nu/2) = \sqrt{[(1+e)/(1-e)]} \cdot \tan(E/2)$$

Worksheet

Use one row per Newton-Raphson iteration. Add rows as needed.

Example 1 ($e = 0.100$, $M = 30.000^\circ$)

n	$E_n (^\circ)$	$\sin(E_n)$	$\cos(E_n)$	$g(E_n)$	$g'(E_n)$	ΔE
0	30.000					
1						
2						

Converged $E = \underline{\hspace{2cm}}^\circ \rightarrow \nu = \underline{\hspace{2cm}}^\circ$

Example 2 ($e = 0.400$, $M = 45.000^\circ$)

n	$E_n (^\circ)$	$\sin(E_n)$	$\cos(E_n)$	$g(E_n)$	$g'(E_n)$	ΔE
0	45.000					
1						
2						
3						

Converged $E = \underline{\hspace{2cm}}^\circ \rightarrow \nu = \underline{\hspace{2cm}}^\circ$